Ex.No:4 Setup and Configure a LAN using a switch and ethernet cables in your lab.

Aim:

To design, set-up and configure a local area network connecting four computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource/data sharing across the network.

Equipment required:

Equipment	Quantity.	specification
PC	4	Ethernet (NIC) enabled.
Network switch	1	8/16/24 port, unmanged / managed.
Ethernet cables (Cat 5e / cat 6)	4	Straight-through, RJ-45 connectors.

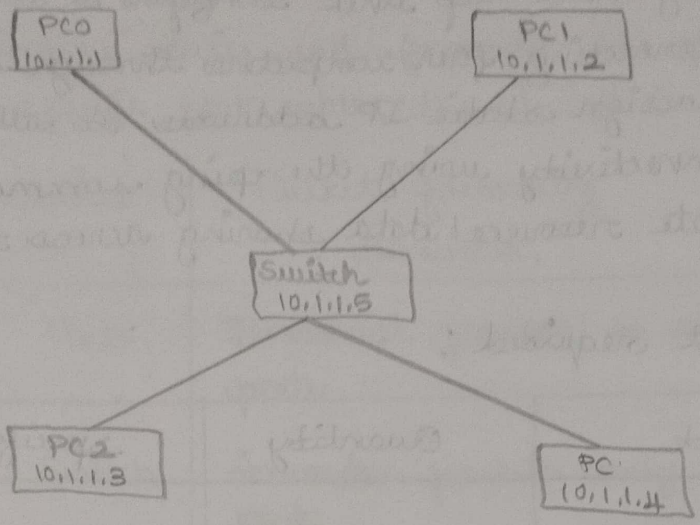
Theory:

A local area network is a network that connects computers and devices within a limited geographical area, allowing them to share resources such as files, and internet connections.

A star topology is used in this experiment, where all end devices (PCs) are individually connected to a central network switch. The switch operates at layer 2 of the OSI model and forwards data frames between connected devices on the MAC addresses. Every device on a LAN must be assigned an IP address within the same network range.

Network diagram

The LAN was implemented using a star topology as shown below, with 4 PC's connected individually to a central switch.



IP address configuration table

Device	IP address	subnet mask	Connected to
PC0	10.1.1.1	255.0.0.0	Switch port Fa0/1.
PC1	10.1.1.2	255.0.0.0	switch port Fa0/2.
PC2	10.1.1.3	255.0.0.0	switch port Fa0/3.
PC3	10.1.1.4	255.0.0.0	switch port Fa0/4.
switch	10.1.1.5	255.0.0.0.	VLAN 1 Interface.

Implementation in Cisco Packet Tracer:

Step 1; Place the devices.

1) Open Cisco Packet tracer.

2) From the device category panel, select End devices and drag 4 PCs onto the workspace.

3) Switch and drag 1 switch onto the workspace.

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Step 2: Cable the devices.

4. click connections → select copper straight-through cable.

5. click each PC, then click the switch, and select an available fast ethernet port to complete each connection.

6. Repeat for all four PCs. green link lights confirm a successful physical connection.

Step 3: Assign IP addresses to PCs.

7. click a PC → desktop tab → IP configuration.

8. Enter the static IP address and subnet mask as per the IP table for each PC.

Step 4: Configure the switch management IP.

9. click the switch → CLI tab.

10. Enter the following commands to assign a management IP:

```
enable
configure terminal
interface vlan1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit
```

Step 5: Verify connectivity using ping

11. click a PC → desktop tab → command prompt.

12. Use the ping command to test connectivity to the other PCs.

13. Successful "Reply from...." messages confirm Connectivity.

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Step 6: Demonstrate data sharing (Server simulation)

14) Add a server device (End devices → Server-PT) and connect it to the switch.

15) Assign it an IP address, subnet mask via its Desktop → IP configuration tab.

16) Open the server's services tab and enable the http service.

17) From another PC, open the web browser and enter the server's IP address to load the page, confirming successful data transfer across the LAN.

Step 7: Observe packet flow in simulation mode.

18) Switch to simulation mode.

19) Send a ping or web request between two devices and use the play/capture controls to step through the packet as it travels hop-by-hop through the switch, observing MAC address based forwarding.

Observation:

All four PCs were successfully connected to the switch in a star topology and assigned unique static IP addresses within the same subnet. Ping tests between all devices returned successful replies, confirming end-to-end connectivity. The shared folder/server resource was successfully accessed from other computer on the network, confirming that the LAN was functioning correctly for resource sharing.